



Nuclear magnetic resonance relaxometry to characterise the decomposition degree of peat soils

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ABSTRACT

An adequate response to the ecological challenges associated with the traditional peatland management and corresponding site-specific measures require spatial information on soil properties and functions, most of which are related to the degree of peat decomposition. Our laboratory study tests the expectation that NMR relaxometry provides simple and rapidly available proxies characterising the decomposition state of peat. We observe that the mean NMR relaxation time is correlated with established soil physical parameters quantifying peat decomposition: water absorption index, unit water content, bulk density and von Post index. The higher the decomposition degree, the faster is the NMR relaxation, which mainly results from a decreasing pore space. Correlation maps between the T_1 and T_2 relaxation times identify hydrogel-like effects in weakly decomposed peat that vanishes if the material decomposes under aerobic conditions. T_1/T_2 ratios of more than ten are observed for peat material with cellular components in contrast to earthified topsoil peat with ratios of less than two. Our attempt to transfer the NMR relaxation data to estimates of water retention functions is partially successful. However, our results also indicate that the relaxation mechanisms in peat are not only controlled by pore sizes. We observe increasing surface relaxivities with increasing decomposition degree, which is most likely the result of a chemical transformation of the pore surface that alters its paramagnetic properties. The magnitude of this increase is significantly higher for T_1 than for T_2 , because the interaction of water molecules and pore surface affects the corresponding NMR relaxation mechanisms differently.

1. Introduction

Peatlands are of essential ecological and economic importance. In their natural state, they serve as reservoirs for water and organic carbon and enable, stabilize and preserve important hydrological, biological and chemical soil functions (e.g. Price, 2003; Holden et al., 2004; Rezanezhad et al., 2016; Lennartz and Liu 2019). In the context of climate change and biodiversity conservation, science, politics and society are questioning the traditional agricultural use of peatland or their mining after drainage, which initiates decomposition processes that changes the peat soil functions irreversibly. Around the globe, activities are discussed or already underway to restore peatlands and/or to reform the land use. In any case, precise knowledge of the state of the peat soil is required, and observation methods capable of monitoring its changes over time are needed, preferably those suitable for studying large spatial scales, i.e. in-situ applications with rapid progress in the field or laboratory analyses that enable fast sample throughput.

Due to the loss of water during peatland drainage, the bulk density of

drained peat, naturally around and even below 0.1 g/cm^3 , increases to a multiple of its original value. Porosity and hydraulic conductivity decrease accordingly (e.g. Letts et al., 2000; Grover and Baldock, 2013; Wallor et al., 2018; Liu and Lennartz, 2018). Air penetrates the soil and creates an aerobic microbiological milieu that initialises oxidation processes. As a consequence, the cellular network of the plant residues in the peat soil collapses into fragments, which reduces the number of macropores in favour of microporosity (Rizzuti et al., 2004; Rezanezhad et al., 2016) and causes a subsidence of the affected surface (Price 2003; Holden et al., 2004).

Under natural conditions, the deeper (and older) regions of the peat layer also decompose. However, the corresponding chemical processes take place under anaerobic conditions and are thus much slower than those triggered by oxidation near the surface. Although the long-term effects of the anaerobic decomposition on the hydraulic soil properties are quite similar as described above (Letts et al., 2000; Grover and Baldock, 2013), its chemical implications are not nearly as severe as those at the drained surface. The oxidation of the drained peat emits CO_2

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and releases nutrients such as nitrogen and phosphorus. These become available to the vegetation, but are also transported deeper into the soil or are washed away by surface water (e.g. Holden et al., 2004; Säurich et al., 2019). The stoichiometric ratios of carbon, nitrogen, hydrogen, and oxygen after chemical analysis of peat are frequently used to characterise the decomposition degree of peat soils (e.g. Boguta and Sokolowska, 2014; Leifeld et al., 2020).

The chemical transformations also cause a significant change of the physical properties of the pore surfaces. Original peat material mainly consists of hydrophilic polysaccharides with polar molecular groups, while its aerobic decomposition produces more and more organic macromolecules with less or without polar sites such as alkanes and aromatic hydrocarbons (Schellekens and Buurman, 2011; Serk et al., 2022). This causes a trend towards hydrophobicity and water repellency (Valat et al 1991). The relationships between hydrophobic and hydrophilic conditions provide another basis for assessing the degree of peat decomposition, e.g. the water drop penetration time (Wu et al., 2020) or the water absorption index (WAI, Gawlik 1992; Boguta and Sokolowska, 2014). Boguta and Sokolowska (2014) demonstrate that the water absorption index is strongly related to chemical properties and functional molecular groups of humic acids in peaty-muck soils.

Another widely used classification scheme for the decomposition degree of peat soils is the von-Post index (von Post, 1924), which describes the quality of decomposition using a scale from H1 (fibric material without any indication of decomposition) to H10 (strongest decomposition without any visible plant structures). Schmidt (1992) propose the unit water content (UWC), which had previously been used mainly in soil mechanics (Ohde 1950), as a simple and quickly available indicator for the degree of peat decomposition. Zeitz & Velty (2002) show that soil degradation leads to a considerable decrease in the UWC of strongly earthified peat compared to that of poorly or moderately earthified peat horizons.

Due to its direct sensitivity to the ^1H protons in the pore water molecules and their physicochemical environment, also the method of nuclear magnetic resonance (NMR) relaxometry is expected to provide reliable proxies for the characterisation of peat soils. Corresponding applications are available for usage at the laboratory scale, in boreholes, and from the surface (Müller-Petke and Yaramanci, 2015). The traditional use of NMR in geophysics is focussed on estimating pore size distribution (Jäger et al., 2009; Costabel and Yaramanci, 2013; Meyer et al., 2018), diffusion coefficient (e.g. Müller-Petke and Yaramanci, 2015), and hydraulic conductivity (Knight et al., 2015; Walsh et al., 2013).

Regarding peat soil, only a few studies are reported to date. Early studies demonstrate that different NMR relaxation regimes in peat samples can be distinguished and related to different binding strengths and pore clusters (McBrierty et al., 1999; Sobottka et al., 1996, and Culligan et al., 2001). Schaumann et al. (2013) investigate how different cations affect the ability of soil organic matter to bind water. Jaeger et al. (2010) and Brax et al. (2017) use NMR relaxometry to observe shrinking and swelling processes in peat samples. Moreover, they identify with the help of correlation maps between longitudinal and transverse NMR relaxation that rewetted peat soil can show a similar NMR relaxation behaviour as organic starch gels. A recent study of Mashhadi et al. (2024) shows that the mean transverse relaxation time measured by borehole NMR tends to smaller values, the higher the von-Post index of the investigated peat soil is.

In this paper, we want to investigate if and how the NMR relaxation behaviour of peat is related to its state of decomposition and if this relationship is quantifiable. In doing so, we make the following hypotheses that are tested and assessed:

1. Spin-lattice and spin-spin relaxation time generally drop with increasing decomposition of peat material because of the corresponding decrease of pore sizes due to the disintegration of the cellular network.

2. The ratio of spin-lattice and spin-spin relaxation rates as indication of a possible gel-like effect (Jaeger et al., 2010; Brax et al., 2017; Buchmann and Schaumann, 2017) changes with changing decomposition degree due to a corresponding change of the physico-chemical properties at the pore surfaces.
3. The NMR relaxation time distribution of a peat sample can be transferred to its water retention function (WRF) similar to the corresponding suggestions for mineral soils (Jaeger et al., 2009; Costabel and Yaramanci, 2013; Meyer et al., 2018).

We present data of two different NMR relaxometry application working at different static magnetic fields to identify a possible influence of the corresponding Larmor frequency (Levitt, 2008) on the results, a single-sided NMR device working at 13.1 MHz (Blümich et al., 2008) and a portable NMR core scanner at 0.5 MHz (Keating et al., 2020).

2. Background of nuclear magnetic resonance (NMR) relaxometry

2.1. NMR relaxation and pore space properties

The principle of nuclear magnetic resonance (NMR) exploits the magnetic moment of proton spins that can be stimulated by electromagnetic excitation. In a static magnetic field B_0 , these magnetic moments are aligned with B_0 and sum up to form the spin magnetisation field M_0 . By applying an oscillating external excitation field, M_0 can be forced to tip out of its equilibrium if the excitation frequency matches the resonance frequency of the proton spins, the so-called Larmor frequency ω_L (Levitt, 2008). After shutting off the excitation, the spin system returns to its equilibrium, a process called NMR relaxation. Both M_0 and ω_L linearly increase with increasing primary field B_0 . The linear factor relating ω_L to B_0 , the gyromagnetic ratio γ , is related to the specific nucleus exhibiting the proton spin ($\gamma = 267.5 \cdot 10^6 \text{ rad/(sT)}$ for ^1H). Depending on the molecular environment of the protons, ω_L can exhibit small characteristic variations, which are analysed in NMR spectroscopy to detect and identify specific molecular compositions. In contrast, the method of NMR relaxometry is only focused on the relaxation process of M_0 and provides insights to fluid-dynamical material properties.

The NMR relaxation behaviour of a water-filled porous medium, which is measured as a multi-exponentially decaying signal (Müller-Petke and Yaramanci, 2015), is mainly influenced by the surface-to-volume ratio (S/V) of its pore space. This is the basis for geophysical NMR applications to estimate characteristic pore size clusters and corresponding hydraulic properties of sediments and porous rocks (Müller-Petke and Yaramanci, 2015). In the context of soil science, this principle can be adapted to provide estimations of the soil water retention function (WRF) as a number of studies have shown (e.g., Jäger et al., 2009; Costabel and Yaramanci, 2013; Meyer et al., 2018; Novotny et al., 2023).

We distinguish two different NMR relaxation processes. The so-called spin-lattice or T_1 relaxation is measured in longitudinal direction to B_0 (Levitt, 2008). The T_1 relaxation component in the n -th pore regime with its individual S_n/V_n reads

$$\frac{1}{T_1^n} = \frac{1}{T_{1,bulk}} + \frac{1}{T_{1,surf}^n} = \frac{1}{T_{1,bulk}} + \rho_1 \frac{S_n}{V_n} \quad (1)$$

$T_{1,bulk}$ and $T_{1,surf}^n$ (in s) are the relaxation times related to the pore water and to the pore surface, respectively (Müller-Petke and Yaramanci, 2015). The surface relaxivity ρ_1 (in m/s) describes the paramagnetic nature of the pore surface, which acts as a sink for the spin-magnetic energy and dominates the relaxation process in the most practical cases. When ρ_1 is known – in practice it is found by adequate calibration – the effective size of the n -th pore regime can be estimated by quantifying S_n/V_n for a specific pore model:

$$\frac{S}{V} = \frac{a}{r_{\text{pore}}} \quad (2)$$

For cylindrical and spherical pores with r_{pore} being the corresponding radius, for instance, the geometry factor a is 2 and 3, respectively. In case of planar pore structure $a = 1$.

The second relaxation mechanism is the so-called spin-spin or T_2 relaxation, measured perpendicular to B_0 , for which the same considerations as for T_1 hold regarding the influence of the pore space. However, there is a significant difference to T_1 , which results from the fact that the T_2 component is much more sensitive to heterogeneities in B_0 . Assuming, for instance, a linear gradient in the B_0 field, the T_2 relaxation rate in the n -th pore regime is given by

$$\frac{1}{T_2^n} = \frac{1}{T_{2,\text{bulk}}} + \frac{1}{T_{2,\text{surf}}^n} + \frac{1}{T_{2,\text{diff}}} = \frac{1}{T_{2,\text{bulk}}} + \rho_2 \frac{S_n}{V_n} + \frac{D(\gamma G t_E)^2}{12} \quad (3)$$

with $T_{2,\text{bulk}}$, $T_{2,\text{surf}}^n$, and ρ_2 defined analogously to equation (1) for T_1 . $T_{2,\text{diff}}$ corresponds to an additional relaxation rate, the diffusion relaxation, that takes into account the diffusion coefficient D , the magnetic gradient G , the gyromagnetic ratio γ , and the echo time t_E . The latter is a specific measurement parameter that controls the timing of the T_2 experiment (Levitt, 2008; Müller-Petke and Yaramanci, 2015). The diffusion relaxation stems from the diffusional movement of the excited spins through the gradient of B_0 during the NMR experiment. Due to the corresponding changes in their Larmor frequencies, they lose their coherence and the observed NMR relaxation process is accelerated (Levitt, 2008; Müller-Petke and Yaramanci, 2015). As mentioned before, $T_{2,\text{diff}}$ must be considered when B_0 exhibits significant heterogeneities. This is the case for some specific NMR measurement technologies such as single-sided (Blümich et al., 2008) and borehole (Knight et al., 2015; Walsh et al., 2013) NMR, whereas usual geophysical laboratory equipment is designed such that B_0 is as homogeneous as possible. Consequently, $T_{2,\text{diff}}$ is often neglected in laboratory practice.

However, it should be noted that the magnetic susceptibility contrast between pore fluid and the pore wall also leads to gradients in B_0 at the pore dimension, which might cause a significant $T_{2,\text{diff}}$ rate. This is particularly the case if ferri- and ferromagnetic minerals have formed in the pore space (Keating and Knight, 2007). Therefore, if the estimation of pore sizes from NMR is in the focus, some laboratory studies have preferred T_1 experiments (Pohlmeier et al., 2009; Costabel et al., 2018). However, Novotny et al. (2023) show that the diffusion modes provide the great opportunity of estimating characteristic pore size clusters of soil samples by bypassing the necessity to determine the surface relaxivity. This can be achieved by combining the response signals of specific pulse sequences, which cancels the usual longitudinal relaxation modes in favour of discovering the diffusion modes related to specific pore dimensions (Novotny et al., 2023).

2.2. Frequency dependence of T_1 and T_2

Both the T_1 and the T_2 relaxation processes depend on ω_L (e.g. Packer, 1977; Levitt 2008) and are described in theory by the following equations:

$$\frac{1}{T_1} = 2\Omega \left[\frac{\tau_c}{1 + \omega_L^2 \tau_c^2} + \frac{4\tau_c}{1 + 4\omega_L^2 \tau_c^2} \right] \quad (4)$$

$$\frac{1}{T_2} = \Omega \left[3\tau_c + \frac{5\tau_c}{1 + \omega_L^2 \tau_c^2} + \frac{2\tau_c}{1 + 4\omega_L^2 \tau_c^2} \right] \quad (5)$$

where τ_c is the correlation time characterising the mean time scale of the molecular fluctuations that the spins experience during the NMR experiments. Ω describes the average coupling strength of the spin system's dipolar interactions. In general, τ_c reflects the mobility of the spin-containing water molecules. In a porous medium, it is restricted, for

instance, by pore walls but also by interactions of the water molecules with the polar groups of large organic molecules. As obvious from the above equations, these restrictions have different effects on the T_1 and T_2 relaxation processes. For the theoretical details, the reader is referred to Levitt (2008).

For the analysis reported here, it is worth noting that the frequency dependency (dispersion) is generally stronger for T_1 than for T_2 . Pohlmeier et al. (2009) have demonstrated with field cycling experiments, i. e. NMR experiments with varying B_0 fields, that the specific shape of T_1 dispersion curves provides information about restrictions on molecular diffusion near the pore surface. According to their findings, water in larger pores related to sand particles shows minimum dispersion effects, while the dispersion of water in smaller pores related to silt and clay is much stronger with doubled T_1 rates at $\omega_L = 0.01$ MHz compared to those at 10 MHz. This is because of an increase in τ_c , i.e. stronger restrictions of the diffusional movement in the pore space between the smallest particles. The same phenomenon is known for organic polymers, where even stronger T_1 dispersion effects have been observed (e.g. de Pasquale et al., 2011; Conte et al., 2021). Therefore, we expect a similar dispersion of the T_1 relaxation for peat.

Indeed, the investigations of Jaeger et al. (2010) and Brax et al. (2017) indicate that the NMR relaxation mechanisms in peat material are predominated by a restricted mobility of the water molecules, which is due to their interactions with the polar molecular groups inside the polymeric network. They found that the T_2 relaxation in peat can be up to ten times faster than T_1 , because the increase of τ_c associated with the reduced mobility of the ^1H spins affects T_2 much more than T_1 . This effect is typical for organic hydrogels. Correspondingly, Jaeger et al. (2010) and Brax et al. (2017) suggest T_1/T_2 correlation maps for analysing peat, especially to identify and characterise this gel-like effect. Based on these findings, we hypothesise that the analysis of T_1/T_2 ratios will also provide insights in the structural and chemical changes of the organic matter that peat material experiences during decomposition. As mentioned before, these changes significantly alter the polar properties of the pore surfaces in peat (Schellekens and Buurman, 2011; Serk et al., 2022).

3. Material and methods

3.1. Test site and sampling

The sample material for this study was gathered from an observation area of the State Office for Mining, Energy and Geology of Lower Saxony (LBEG), the Gnarrnburger Moor in the Northwest of Germany, where long-term investigations have been conducted to establish and maintain a sustainable peatland management (Kalinski et al., 2021). The samples were taken on a transect covering the transition of mineral to peat soil on the edge of a former bog land that has been drained decades ago and has since been used as grassland for cattle feeding. Correspondingly, the topsoil consists of strongly decomposed amorphous peat, in which no cellular plant residuals can be found anymore (Fig. 1a). The decomposition degree of the peat material decreases quickly within the following few decimetres (Fig. 1b) until, at a depth of less than 0.5 m, almost non-decomposed peat with clearly visible cellular structure is present (Fig. 1c). Beneath this gradual zone, there is a sequence of fibric (white) peat and sapric (black) peat with varying thicknesses, a situation that is representative for most of the bog lands in Northern Germany. Below the peat body, clayey and loamy sand has been found (Fig. 1d). From now on, we use the terms 'white' and 'black' peat according to the instructions for peat soil mapping of Lower Saxony, Germany (Caspers, 2010) as synonyms for 'fibric' and 'sapric'. In doing so, we follow the principle characterisation suggested by Taufic et al., 2019, who relate the different von Post degrees to the categories of fibric (H1-H3), hemic (H4-H6), and sapric (H7-H10) peat. While hemic peat is underrepresented and negligible in our study area, the white peat can clearly be identified by sphagnum moss residuals, whereas in the black peat the

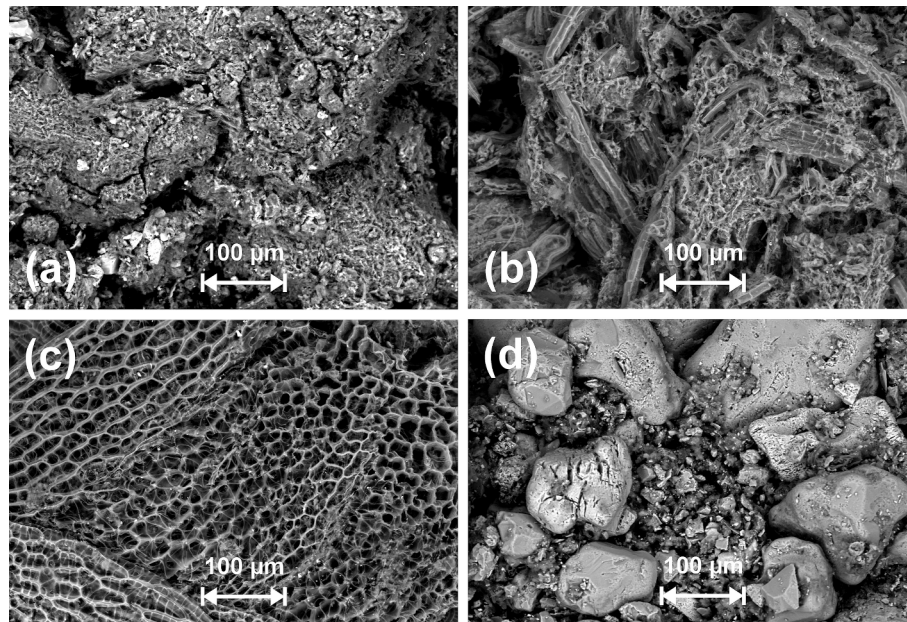


Fig. 1. Pictures from scanning electron microscope: (a) amorphous topsoil peat (von Post index: H10, position: P2, depth: 0–5 cm), (b) moderately decomposed peat (von Post index: H8, position: P1, depth: 15–20 cm), (c) weakly decomposed (white) peat (von Post index: H2, position: P2, depth: 38–43 cm), (d) mineral soil below the peat (position: P1, depth: 61–66 cm).

sphagnum leaves are completely disintegrated and decomposed to form an amorphous muddy matrix. However, the cellular structure of other plant residuals such as cottonsedge and horsetail are still visible in black peat material. Chemical analyses have shown that the contents of C_{org} and N in the white peat layer varies only little in the range 45–50 % and around 1 %, respectively. In the topsoil, the C_{org} content decreases to values lower than 40 % down to a minimum of 27 %, while the N content approximately doubles compared to the undecomposed original material.

We show data from two different sampling campaigns. The first set of samples (Set A) was taken during the initial exploration of the area in 2022. At seven points along the transect (positions P1 to P7), a gouge auger with a diameter of 2.5 cm was used to identify the peat basis and the soil stratigraphy (Fig. 2a). For each characteristic soil layer, the associated material was removed from the slotted probe and, for a small portion of it, the von Post index was determined. The residual material was packed in a plastic bag for later analyses in the laboratory. A small amount of this material (about 40 ml) was prepared for measurements using a single-sided NMR device (Fig. 2b and c).

In 2023, four sampling sites were determined (Set B), three inside (P1, P2, and P4) and one outside of the peat area (P0), where undisturbed samples were taken from two different depths levels (0.1 m and 0.5 m). For each sampling site and depth level, one subsample was prepared using a standard ring (250 ml, stainless steel), and three further subsamples using a plastic sample holder with a volume 150 ml (Fig. 2d). The latter were necessary for the NMR measurements that must not have any metal inside the sensitive measurement range to ensure a reliable electromagnetic signal detection. In addition to the undisturbed samples, we gathered additional sample material using a gouge auger with a diameter 5 cm (Fig. 2e) to get a larger amount of sample material also from deeper regions. This material was repacked in the same sample holders as the undisturbed samples. The undisturbed and the repacked samples were analysed using the two aforementioned NMR devices, i.e. the single-sided apparatus and the mobile scanner (Fig. 2f). Additional material was taken from each sampling position of Set B and stored in plastic bags as a reserve for further analyses.

3.2. Soil physical reference measurements

The decomposition degree of peat is most often determined using the scheme of von Post (1924). Although non-quantitative and subjective, it is a practical qualitative measure for rapid field use. However, for the quantitative comparison with the NMR measurements we prefer rather objective reference quantities to characterise the degree of decomposition such as the water absorption index (Gawlik, 1992; Boguta and Sokolowska, 2014), and the unit water content (Ohde, 1950; Schmidt, 1992), in the following referred to as WAI and UWC, respectively. As shown for instance by Grover and Baldock (2013), Wallor et al. (2018), Liu and Lennartz (2018), among many others, also the bulk density (BD) is an strong indicator for the decomposition of peat, which is also tested in this study against the measured NMR relaxation parameters.

Gawlik (1992) suggested an easily available measure to characterise the state of peat decomposition, which compares the water holding capacity of the original material with its at 105 °C dried representative. The drying process causes a variety of changes in the samples. Two of the most important are the increase in density and the change in surface chemistry which results in stronger and often prolonged inhibition of wetting. In order to make the WAI measurements more practicable and to enable a higher sample throughput, the original procedure of Gawlik (1992) was adapted to 50 ml centrifuge tubes. Each peat sample was divided into 5 specimens. The field-moist material was filled into the inlet of the centrifuge tube in several portions and compacted with a plunger at approx. 25 kPa for each portion. The inlet was filled to a height of 2 cm. The added amount of wet peat was determined by weighing and converted to dry mass using the previously performed dry mass determination (oven-dried for 24 h at 105 °C) for the sample under study. Also five new inlets were prepared with the dried peat such that, related to the dry mass, the dry and wet specimens exhibit the same amount of material. Compaction was not necessary for the dry specimens, because the filling height did not exceed 2 cm. The filled inlets were placed in a glass dish filled with water for 7 days. To prevent the peat material from floating, the specimens were weighted down with 20 ml glass vials. The water level in the glass dishes was controlled daily and adjusted to approximately 1 mm above the sample surface. After a week of saturation, all samples were centrifuged with 3000 rpm (which corresponds to a pore pressure of about 100 hPa) for one hour and

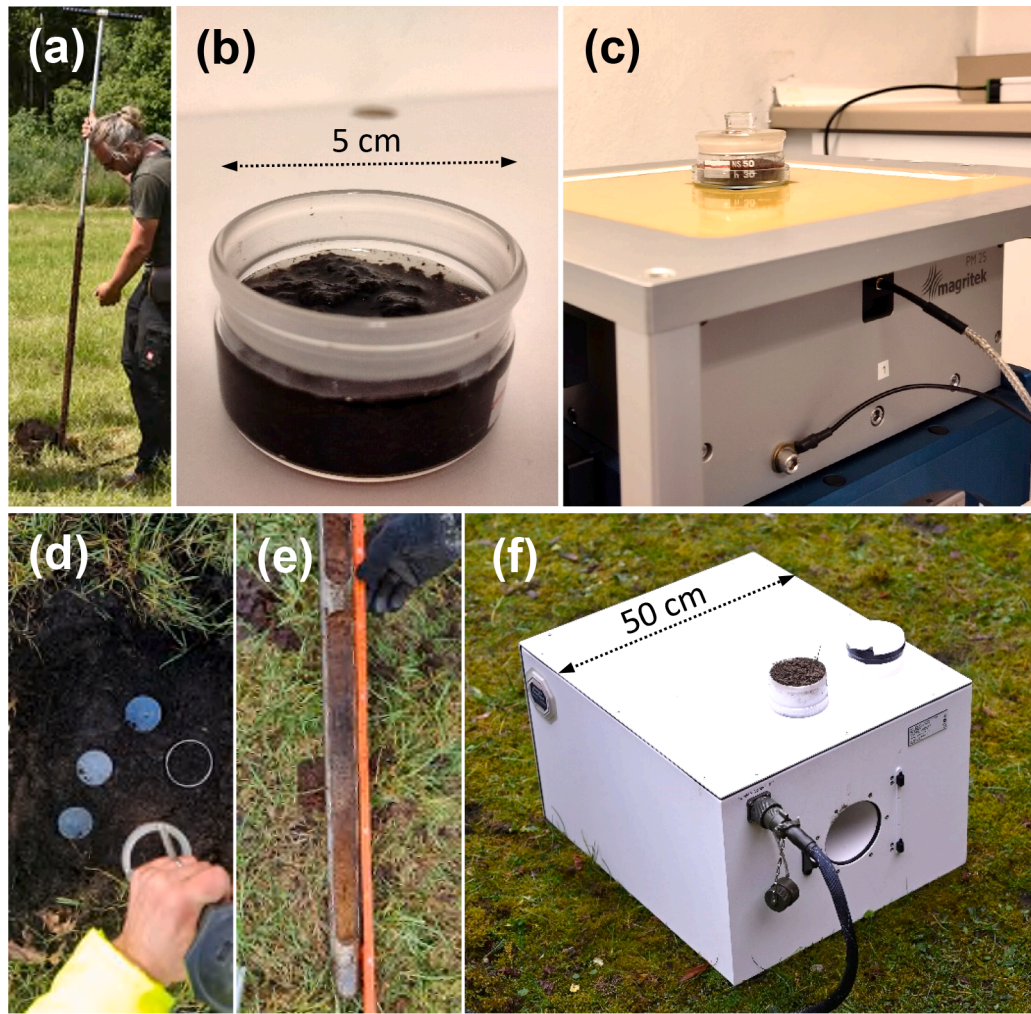


Fig. 2. Two sampling campaigns for NMR measurements were conducted: (top) peat material from a 2.5 cm gouge auger (a) was repacked in sealable glass jars (b) and investigated using a table-top single-sided NMR device (c), (bottom) undisturbed samples in plastic sampling rings up to a depth of ~ 0.5 m were taken (d), material from deeper regions > 0.5 m was obtained using a 5 cm gouge auger (e) and repacked in plastic sampling rings for NMR measurements using a mobile core scanner (f).

weighed to determine the water content. Finally, the WAI was calculated by dividing the water content of the dried and rewetted samples (θ_{rewet}) by the water content of the saturated original samples (θ_{orig}):

$$WAI = \frac{\theta_{\text{rewet}}}{\theta_{\text{orig}}} \quad (6)$$

The UWC is a parameter developed in soil mechanics to describe the strength of soils (Ohde 1950). Schmidt (1992) transferred the method to characterise peat soils and developed a framework for estimating the decomposition degree of peat. Since the UWI determination in our study strictly follows the procedure of Schmidt (1986), it is described here only briefly. The peat material is prepared with distilled water to form a paste-like sample and placed in a pot and gradually loaded up to a standard load of 100 kPa. The pore water can escape through a porous filter stone for 24 h. After 24 h, the leaked water was completely removed and the sample was relieved. The moisture resulting from the load is determined as the gravimetric water content:

$$UWC = \frac{m_{\text{load}} - m_{\text{dry}}}{m_{\text{dry}}} \quad (7)$$

with m_{load} and m_{dry} being the masses of the loaded and the at 105 °C dried sample, respectively.

The BD was determined after the NMR experiments, when the

samples of set A (repacked) and B (undisturbed) were oven-dried at 105 °C.

In addition to the aforementioned soil physical quantities, the water retention function (WRF) for the samples of Set B was measured using the evaporation method (Weber et al., 2017a). The WRF is frequently interpreted as cumulative pore size distribution (e.g. Weber et al., 2017a, 2017b). The equivalent effective pore radius r_{eff} and the corresponding capillary pressure head h are related to each other by the Young-Laplace equation (Weber et al., 2017b):

$$|h| = \frac{2\sigma\cos(\beta)}{g\rho_w r_{\text{eff}}} \quad (8)$$

where σ is the surface tension of the pore water, β it's contact angle with the pore surface, g the gravitational acceleration, and ρ_w the water density. Weber et al. (2017a, 2017b) identify three different pore clusters in living sphagnum moss, inter-, intra- and inner-plant, corresponding to the $|h|$ -ranges < 10 cm, 10–100 cm, and > 100 cm, respectively. Moreover, they show that, due to the starting decomposition and compaction, the sphagnum peat soil beneath the living sphagnum layers does not exhibit inter-plant pore space.

Because the T_1 and T_2 surface relaxation rates (equations (1) and (3)) and h are related to the pore size, the following relation is anticipated:

$$\frac{1}{T_{1/2,surf}} = C_{1/2} \cdot |h| \text{ with } C_{1/2} = \frac{g\rho_{pw}}{2\sigma\cos(\beta)} a\rho_{1/2} \quad (9)$$

We anticipate $g = 9.81 \text{ m/s}^2$, $\rho_w = 1000 \text{ kg/m}^3$, $\beta = 0$ (i.e. perfect hydrophilic properties) and $\sigma = 73 \times 10^{-3} \text{ N/m}$ corresponding to a room temperature of 20°C that prevailed in the laboratory during the experiments.

3.3. NMR measurements

3.3.1. Mobile NMR technology

The NMR analyses were conducted using mobile measurement equipment (Fig. 2c and f). The application of these devices is not limited to specific laboratory conditions. Even a usage in the field is possible. Moreover, another specific advantage of the single-sided NMR technology is the little amount of sample material that is required. A small coil with the size of a few square centimetres is installed on an array of permanent magnets (Blümich et al., 2008). The investigated sample material (Set A) was repacked in sealable glass jars of 40 ml (Fig. 2c), compressed with a plunger at 25 kPa as the sample material for the reference measurements (Section 3.2), saturated with distilled water, and put on top of the device. The sensitive footprint of the single-sided NMR measurement lies inside the sample and is a thin section (thickness: 200 μm) parallel to the surface of the magnets with lateral dimensions similar to the measurement coil. We used a PM25 measurement system (Magritek) with a working frequency of 13.1 MHz. T_1 and T_2 measurements are conducted using the saturation recovery (SR) and the Carr-Purcell-Meiboom-Gill (CPMG) sequences (Levitt, 2008; Müller-Petke and Yaramanci, 2015), respectively. The echo time was 68 μs and the number of measurement repetitions (scans) for averaging was varied between 4 and 64 to achieve signal-to-noise ratios (SNR) of at least 40. The B_0 gradient inside the sensitive volume causes a $T_{2,diff}$ of 0.19 s, which was included in the data interpretation according to the suggestion of Costabel et al. (2023).

The used mobile NMR core scanner (Helios, Vista Clara) allows for measuring cylindrical samples or drilling cores immediately after taking them in the field. The sensitive volume inside the device is several hundred ml. The samples of Set B (150 ml plastic sample holders, see section 3.1) were placed in the middle of the sensitive volume, so that the NMR signal represents the sample material as a whole. The beneficial combination of transportability of the device and analysis of large sample volumes comes at the price of a lower magnetic field, because the installed permanent magnets have to be reduced in size and weight compared to usual laboratory NMR equipment. The corresponding operating frequency is 0.5 MHz. T_2 relaxation measurements are conducted using the CPMG sequence with an echo time of 200 μs . T_1 is measured by inversion recovery (IR, Levitt, 2008; Müller-Petke and Yaramanci, 2015). For the peat samples, we chose numbers of scans from 16 to 128 to get SNRs of at least 30. To achieve a similar range for the mineral soils samples with much lower water contents, the number of scans had to be set to 512–1024. All samples were saturated with distilled water before the NMR relaxometry measurements.

For both NMR applications, a sample with degassed water (= 100 % water content) was used to transfer the measured signal amplitudes to absolute measures of volumetric water content. The relaxation time distributions (RTD) of both T_1 and T_2 measurements were calculated using the open source software NUCLEUS (Hiller, 2024). In doing so, the bulk relaxation time (see equation (1) and (3)) was neglected. We extrude the pore fluid from a portion of the retained sample material by squeezing and determined $T_{1/2,bulk}$ values around 1.5 s, which is long enough to verify the assumption that the influence of the bulk relaxation is insignificant. For the comparison with the WRF, the RTDs were cumulatively summarized. A brute force parameter search was applied to get a best fit of WRF and cumulative RTD (equation (9)) following the least square criterion.

3.3.2. Combined measurements of longitudinal and transverse NMR relaxation

As described in Section 2.2, 2D correlation maps between T_1 and T_2 can provide useful information about the mobility of water molecules in peat. Thus, we combined the pulse sequences of CPMG with those of SR (single-sided NMR) and IR (NMR-Scanner), which leads to the following equations for quantifying the evolution of the measured signals V_{SR} and V_{IR} (Song et al., 2002):

$$V_{SR}(\Delta, t) = \iint \left(1 - e^{-\frac{\Delta}{T_1}}\right) e^{-\frac{t}{T_2}} I(T_1, T_2) dT_1 dT_2 \quad (10)$$

$$V_{IR}(\Delta, t) = \iint \left(1 - 2e^{-\frac{\Delta}{T_1}}\right) e^{-\frac{t}{T_2}} I(T_1, T_2) dT_1 dT_2 \quad (11)$$

V_{SR} and V_{IR} are functions of the recovery time Δ and the CPMG measurement time t . $I(T_1, T_2)$ is the 2D intensity distribution in the $T_1 - T_2$ domain. The latter is reconstructed using a 2D Laplace inversion algorithm (Song et al., 2002) that was realised with the Matlab-based software NUCLEUS (Hiller, 2024).

4. Results and discussion

4.1. Characterisation of the peat decomposition along the test profile

As stated earlier, the investigation site exhibits a soil sequence representative for many former bog regions that have been drained and used for agriculture in Northern Germany. Fig. 3 shows the sampling points covering the transition from the surrounding mineral soil (P0) onto the peat body (P1 to P7). The general sequence consists of a highly decomposed and amorphous topsoil peat (von Post index: H10) with a thickness of 0.1 m to 0.2 m, a white peat layer with thicknesses of 1.0 m to 1.5 m, and a final black peat layer down to depths of at maximum 5 m. The white peat is characterised with low von Post indices from H1 to at maximum H4, whereas for the black peat layer, the index varies between H6 and H9.

The varying decomposition degrees of the different peat layers are well represented by the WAI and the UWC in Fig. 3, where the two quantities show reversed trends with depth. The topsoil is characterised high WAIs of > 0.4 and low UWCs < 2 . The subsequent white peat layer below the topsoil horizon, i.e. below the region with frequent agricultural processing, is indicated by a remarkable jump of the two quantities, the WAI towards lower and UWC towards higher values. On the most sampling positions, the WAI in the white peat layer exhibits a gradient towards lower values with increasing depth, from 0.3 to less than 0.2. The corresponding (reverse) trend for the UWC is not that obvious on all positions. Its values in the white peat layer vary between 3 and 6, while its highest values are found at deeper regions. Inside the black peat, WAI and UWC also show some variations without a clear depth-dependency, between 0.1 and 0.2 for the WAI and between 5 and 6 for the UWC. The mineral soil below and outside the peat body shows up with high WAI up to 0.8 and low UWC below 1.

4.2. T_1/T_2 correlation maps

On P4, NMR data was acquired for the complete vertical soil profile using both NMR applications for a detailed comparison of their performances. Fig. 4 shows the corresponding T_1/T_2 correlation maps and their assignment along the depth profile. For the amorphous peat sample from the topsoil, the T_1/T_2 patterns at 13.1 MHz and 0.5 MHz are almost in exact agreement. The signal intensity distribution in the $T_1 - T_2$ domain is very close to the 1:1 line meaning that the low T_1/T_2 ratio (< 2) is almost the same for all pore clusters. The covered relaxation times range between 0.003 s and 0.1 s. For the deeper peat layers, the average T_1/T_2 ratios are higher with values from 4.13 to 11.66 and 2.44 to 4.00

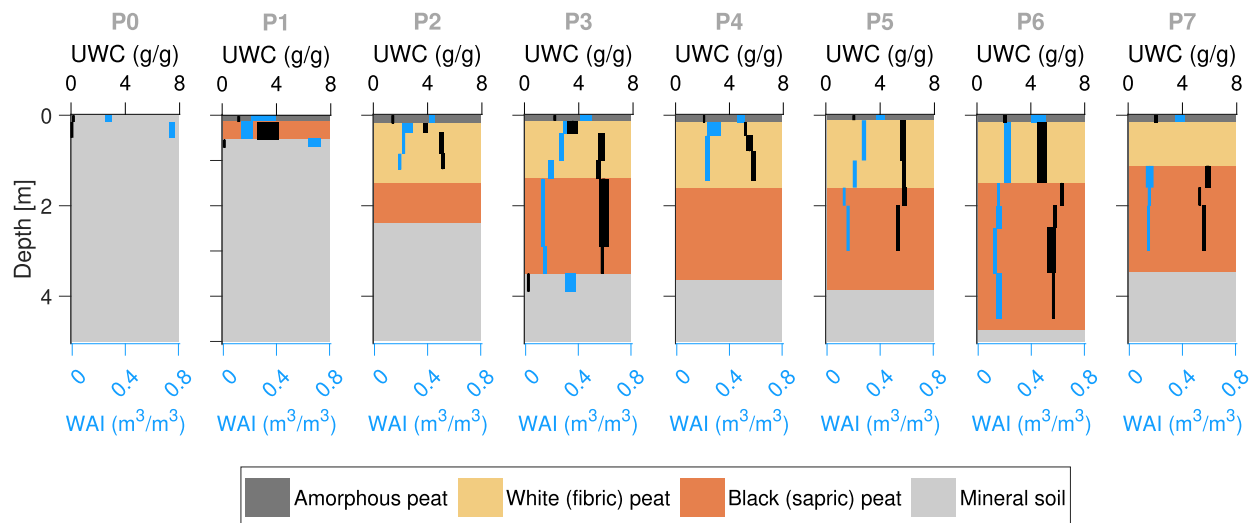


Fig. 3. Soil profiles along the investigated transect with corresponding distributions of water absorption index (WAI) and unit water content (UWC), which serve as reference data in this study representing the state of decomposition.

for the measurements at 13.1 and 0.5 MHz, respectively. The distribution deviates more and more from the 1:1 line, the smaller the relaxation times are. The T_2 distributions of the white peat ranges from about 0.01 to 0.1 s, whereas those of the black peat are broader between 0.005 to 0.5 s. Although the principle T_1/T_2 patterns at 13.1 MHz and at 0.5 MHz are very similar for each individual sample, the T_1/T_2 ratios measured at 13.1 MHz are generally at least two times larger.

In agreement with our observations, Jaeger et al. (2010) also report that the T_1/T_2 ratios in their peat samples increase with decreasing relaxation times. They explained this phenomenon by a restricted rotational mobility of the pore water molecules in the vicinity of the organic pore walls. These restrictions increase with decreasing pore size. According to the conceptual suggestions of Brax et al. (2017) and Buchmann and Schaumann (2017), a certain amount of the water is entrapped in a gel-like phase, i.e., interactions (hydrogen bonds) between the pore water molecules and the polar molecular groups of the organic matter. The corresponding binding forces are weaker than, for instance, capillary or absorption forces, which have a very similar impact on both T_1 and T_2 NMR relaxation processes. Instead, in the presence of organic pore surfaces with polar properties, a restricted rotational mobility will accelerate the T_2 relaxation much more than T_1 . The same effect is reported for bio-polymers and starches (Packer, 1977; McBrierty et al., 1999; Hill et al., 2005; Brax et al., 2019).

However, the structural breakdown of the fibric peat matter in the topsoil during the aerobic decomposition has two obvious effects on the measured NMR relaxation. First, it causes a decrease of the T_1/T_2 ratio, because the chemical transformation of the organic molecules leads to a decreased number of polar sites on the pore surface. Second, the overall relaxation process (both T_1 and T_2) is faster, which indicates a general decrease of the pore dimension. If the peat material is transformed under anaerobic conditions (black peat), the cellular structure only partially disintegrates. We hypothesize that the observed broadening of the T_1 and T_2 distributions of the black peat indicates the corresponding appearance of smaller pores, however, at the same time the pore space opens in parts due to the disintegration, which is indicated by the occurrence of longer T_1 and T_2 times in the distributions compared to the white peat. Interestingly, the principle hydrogel properties of the peat remain intact, which is indicated by the T_1/T_2 ratio of the black peat that is only slightly smaller than that of the original white peat.

These qualitative observations are made for both NMR measurement application. However, the NMR scanner operating at 0.5 MHz measures lower T_1/T_2 ratios and thus lower contrasts between different decomposition states. This trend was expected beforehand, as described earlier

in section 2.2, because organic matter is reported in the relevant literature to cause a significant frequency dispersion of the T_1 relaxation (de Pasquale et al., 2011; Conte et al., 2021), i.e. the T_1 relaxation time increases with increasing frequency much stronger than T_2 (Packer, 1977; Levitt, 2008).

4.3. Correlation of mean relaxation times with peat decomposition degree

In the previous section, we demonstrated with an example soil profile that the NMR relaxation behaviour of the different peat layers exhibits characteristic patterns. In this section, we want to investigate if a simple average value of the NMR relaxation distribution measured in peat can serve as a suitable measure for assessing its decomposition state. The logarithmic mean of the T_1 and T_2 relaxation time distributions ($T_{1,lgm}$ and $T_{2,lgm}$) has been proven to be a powerful quantity for estimating the hydraulic conductivity and/or effective pore size of mineral sediments (Walsh et al., 2013; Knight et al., 2015; Müller-Petke and Yaramanci, 2015). Thus, we also make use of it here for our peat material. Fig. 5 (13.1 MHz) and Fig. 6 (0.5 MHz) correlate the $T_{1,lgm}$ and $T_{2,lgm}$ times, as well as the ratio of the two, with the corresponding reference values (WAI, UWC, BD, von Post index) for all of our investigated samples. Obviously, the relaxation times are negatively correlated with the WAI and positively with the UWC. Corresponding linear regressions in the semi-logarithmic domains leads to R^2 values between 0.67 and 0.83. The R^2 values for the positive correlations with BD and the von Post index are smaller with 0.33–0.68 and 0.26–0.31, respectively. We observe that the slope of the regression lines for all reference values is different for $T_{1,lgm}$ and $T_{2,lgm}$ with generally smaller slopes for $T_{2,lgm}$. Correspondingly, the ratio $T_{1,lgm}/T_{2,lgm}$ must also be correlated with the respective reference quantities (subfigures c, f, j, and m in Figs. 5 and 6). However, the corresponding R^2 values are smaller than those for $T_{1,lgm}$ and $T_{2,lgm}$. For both NMR applications in use, the correlation of the $T_{1,lgm}/T_{2,lgm}$ ratio with the UWC is the strongest.

Our observation that the logarithmic mean relaxation times decrease with increasing decomposition, which corresponds well to the findings of a recent borehole NMR study from Denmark (Mashhadi et al., 2024), verifies our initial expectations (hypothesis 1). Due to the disintegration of the cellular structure during decomposition, the overall pore dimension decreases and with it the mean relaxation times. The R^2 values for the correlations of $T_{1,lgm}$ and $T_{2,lgm}$ with the well-established von Post index appear to be weaker compared to those with WAI and UWC, which can be explained by the fact that this parameter must, strictly speaking, be treated as a rather qualitative measure and the transfer of the index

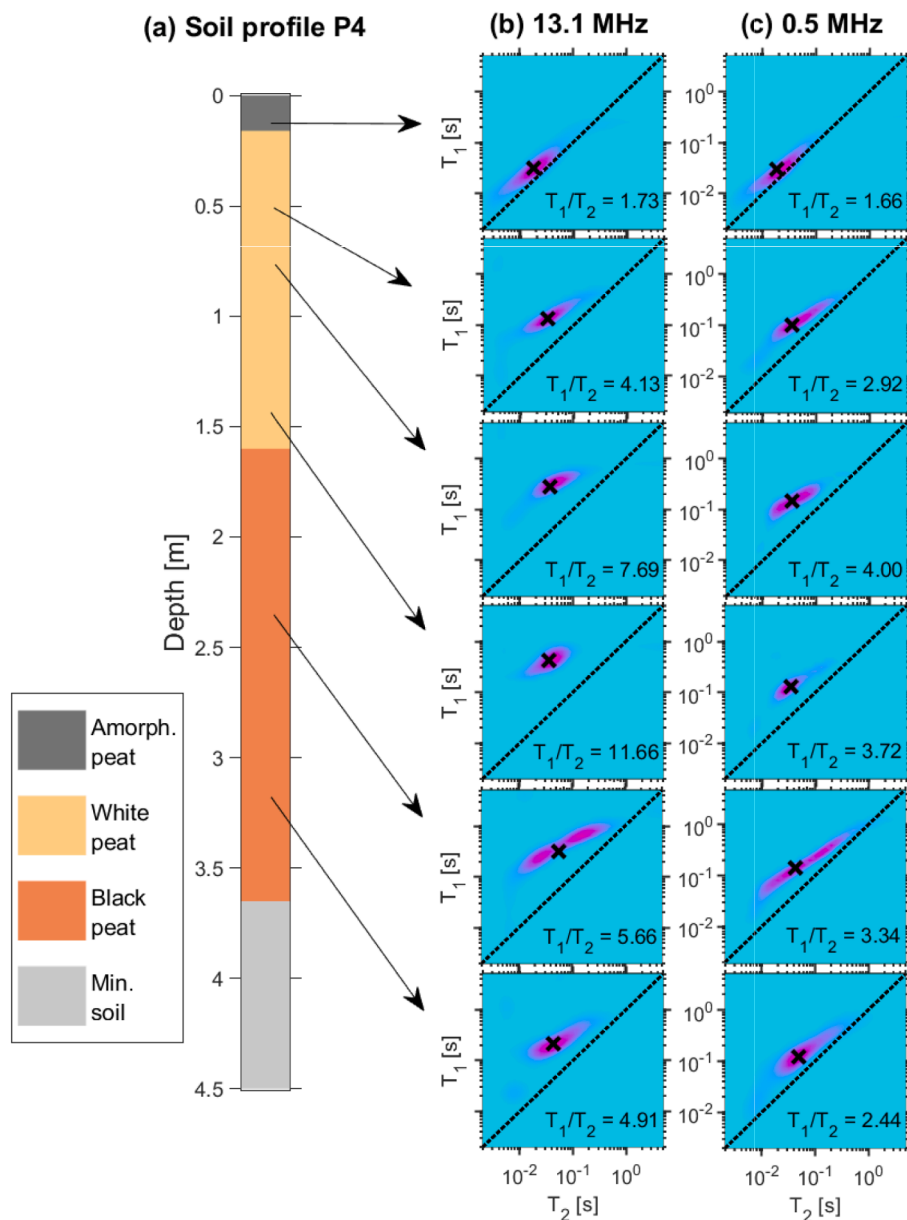


Fig. 4. (a) Soil profile on sampling point P4 and T_1/T_2 correlation maps as measured for diverse samples along the profile using the single-sided NMR (b) and the mobile NMR core scanner (c).

for a quantitative analysis is not optimal.

Moreover, we interpret the fact that also the $T_{1,lgm}/T_{2,lgm}$ ratio is correlated with the decomposition degree to be a strong indication that it is not the decreased pore dimension alone that affects the NMR relaxation. A decreasing pore size has the same quantitative effect on both T_1 and T_2 relaxation (see equations (1) to (3), i.e. the ratio between $T_{1,lgm}$ and $T_{2,lgm}$ should be constant for all peat decomposition states. This is obviously not the case, which can be explained by an additional alteration of the surface relaxivity that develops differently for ρ_1 and ρ_2 during decomposition. We hypothesise that the reason for this development is the transformation of the original peat material into molecules with less polar sites, which leads to an equalization of the T_1 and T_2 relaxation rates. As explained in section 2.2, the less-decomposed molecules in the intact cellular structure of the original peat has many polar sites. The ^1H spins-bearing water molecules are trapped at the pore surface and cause a T_2 relaxation much faster than T_1 . In other words, the hydrogel effect (Jaeger et al., 2010; Brax et al., 2017; Buchmann and Schaumann, 2017) can be observed distinctly for weakly decomposed

peat and vanishes for strongly decomposed amorphous peat, which verifies our second hypothesis. We also hypothesise that this is the reason for the weaker correlation of the NMR parameters with the BD values. In contrast to WAI and UWC, which indicates the physicochemical alteration of the pore surfaces to some extent, the BD is insensitive to it.

It has to be noted that the WAI as a reliable measure for peat decomposition is still a matter of scientific discussion. As previously explained, many properties of the sample material change during the drying process. In addition to the total pore volume and physicochemical surface properties such as wetting, also the pore structure can be significantly and permanently altered. Further research is necessary on this issue, irrespective of the NMR aspects discussed in this paper.

4.4. NMR relaxation time distribution (RTD) and water retention function (WRF)

Both RTD and WRF are related to the pore size distribution (section

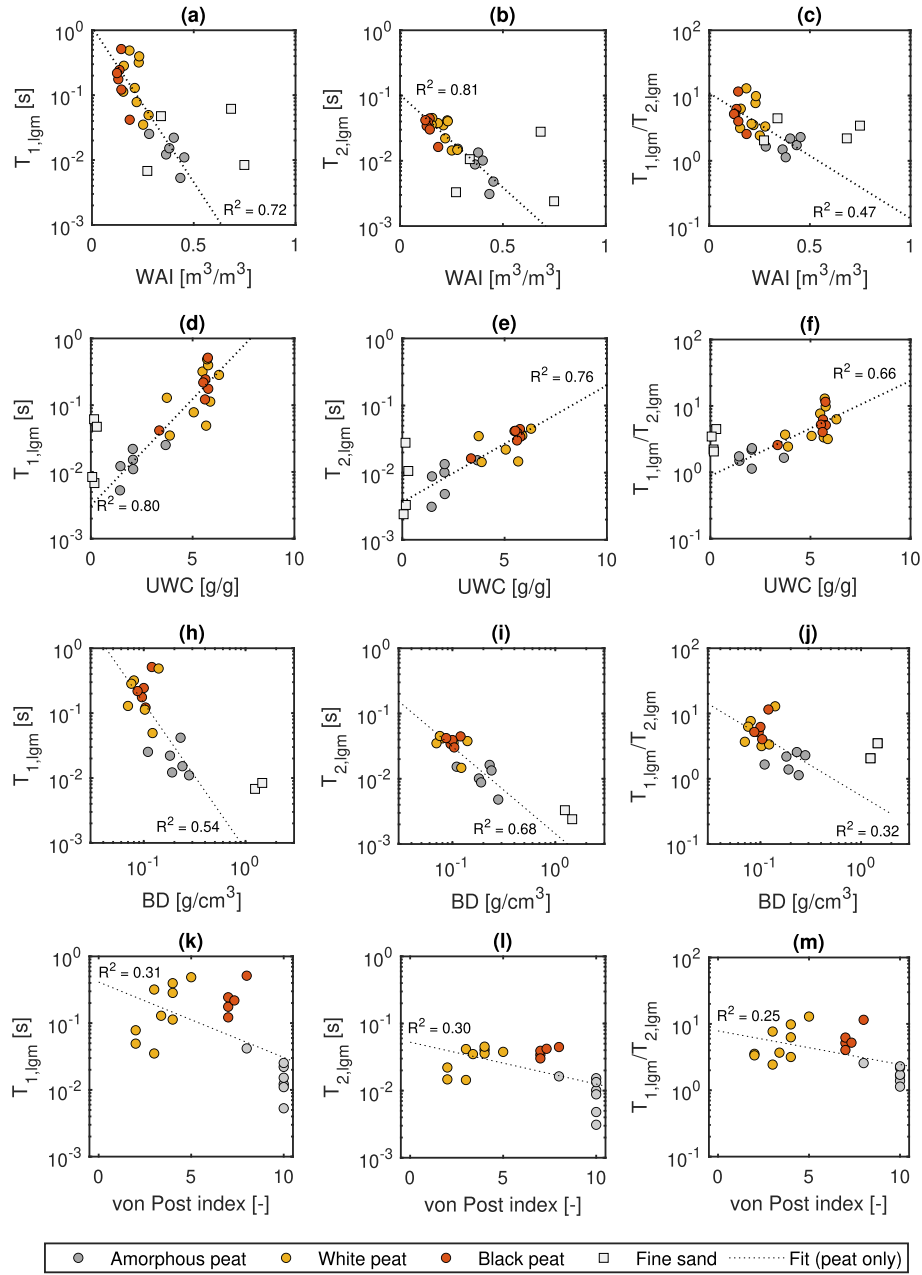


Fig. 5. Crossplots of mean logarithmic relaxation times $T_{1,lgm}$ and $T_{2,lgm}$ as measured using the single-sided NMR (13.1 MHz) and the water absorption index WAI (a to c), the unit water content UWC (d to f), the bulk density BD (h to j), and the von Post index (k to m). The linear regression lines are calculated in the semi-logarithmic domain for the data of the peat soil (circles), i.e. without considering the data of the mineral soil (unfilled squares).

2.1 and 3.2) and one might expect the two to be linked to each other. For the case that such a link existed, a potential opportunity opens up: the WRF of peat soil samples could be provided much faster with a much higher sample throughput than by using conventional WRF measurement techniques. However, in the context of peat soils, it is unknown to date whether or not a corresponding link between RTD and WRF actually exists. On the one hand, it is unknown how the pore geometry (and correspondingly the geometry factor a in equation (2)) exactly changes during decomposition. For instance, one might expect the cellular polymer network of less or non-decomposed peat, i.e. the inner-plant pore space according to Weber et al., 2017a, to be dominated by more or less spherical pores. During decomposition and the corresponding breakdown of the cellular structure, the principal pore geometry obviously changes and even planar pores might appear between the remaining fragments. The consequence regarding the NMR relaxation

behaviour would be a corresponding change in the parameter a (equation (2)) from 3 to 1. On the other hand, it is unclear whether the surface relaxivity $\rho_{1/2}$ (equation (1) and (3)) can be treated as an effective material constant. If its variability inside the probed peat sample is low enough, its effect on the NMR relaxation behaviour is similar on average throughout the entire observed pore space. To date, there is no information in the literature on how heterogeneous the distribution of $\rho_{1/2}$ in peat might be, neither among different types of peat soils nor among the pores in the very same sample.

Although these issues remain important objectives for future research to improve the interpretation of NMR relaxometry data in the peat context, we want to empirically compare the example WRFs measured in this study with the cumulative RTDs of the corresponding reference samples (Fig. 7). In doing so, we transfer the T_1 and T_2 times to corresponding representations of $|h|$ following the considerations

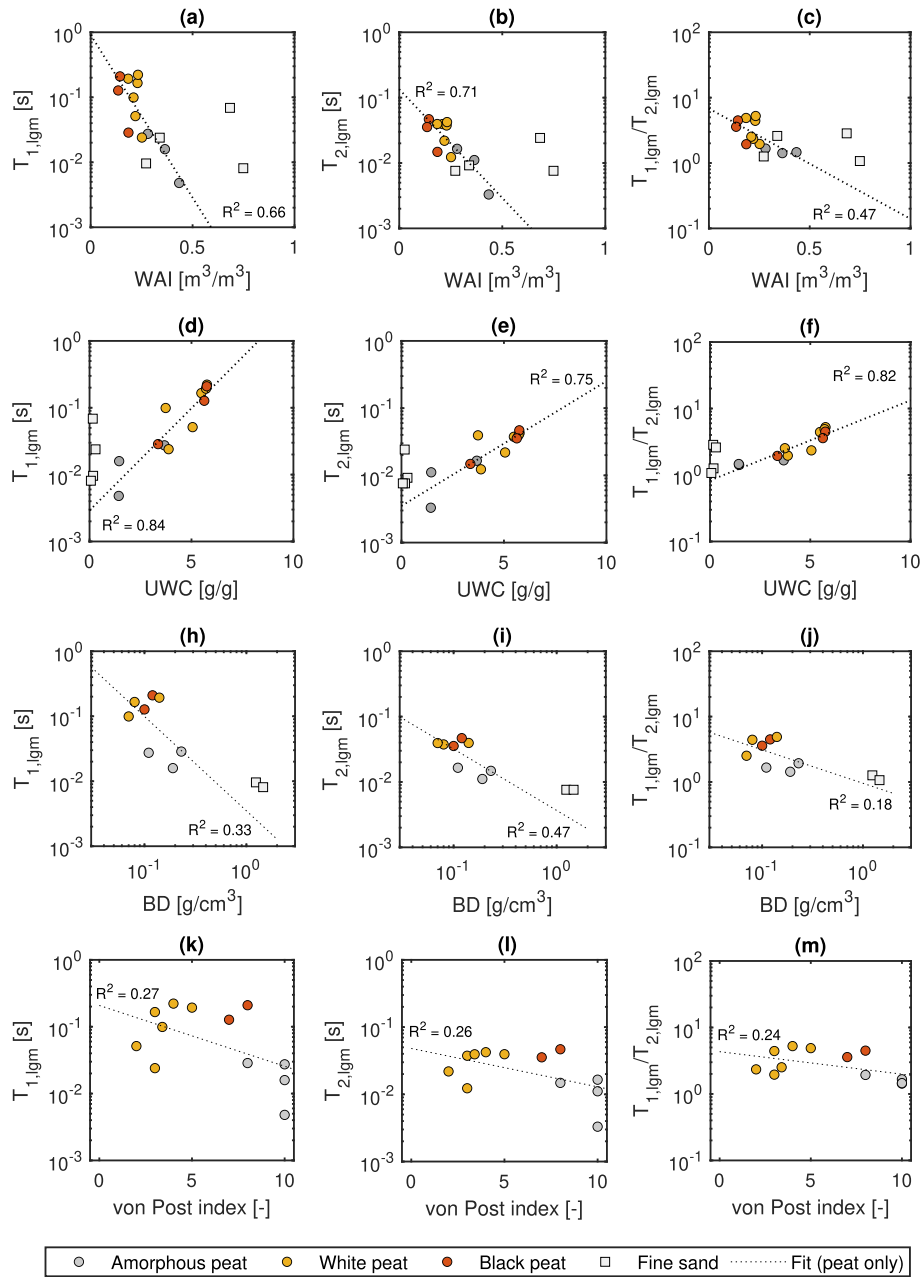


Fig. 6. Crossplots of mean logarithmic relaxation times $T_{1,lgm}$ and $T_{2,lgm}$ as measured using the NMR core scanner (0.5 MHz) and the water absorption index WAI (a to c), the unit water content UWC (d to f), the bulk density BD (h to j), and the von Post index (k to m). The linear regression lines are calculated in the semi-logarithmic domain for the data of the peat soil (circles), i.e. without considering the data of the mineral soil (unfilled squares).

represented by equation (9). The constant $C_{1,2}$ is determined such that the cumulative RTD matches the WRF according to the least square criterion. From the results of this fitting process and assuming $g = 9.81 \text{ m/s}^2$, $\rho_{pw} = 1000 \text{ kg/m}^3$, $\beta = 0$, and $\sigma = 73 \times 10^{-3} \text{ N/m}$, we calculate estimates for the term $a \cdot \rho_{1/2}$ (see Table 1). As mentioned before, we do not know exactly, what geometry factors (a) might represent the pore space of peat at the different decomposition states. However, by using the estimates of $a \cdot \rho_{1/2}$ we attempt to observe and assess how both pore space geometry and surface relaxivity may vary for the different kinds of peat existing in our study area.

We observe in Fig. 7 that the results of the two NMR methods are very similar to each other for every sample. The shapes of the T_2 -RTDs match the WRF measurements better than those of T_1 . For the latter, the WRFs generally exhibit a smoother slope than the RTDs, which is also the trend for the T_2 data at the depth ranges 1.30–1.60 m and 3.1–3.5 m.

Following the assumption of comparing cumulative pore size distributions in Fig. 7, the RTDs seem to underestimate their widths. Weber et al. (2017a, 2017b) show a bimodal distribution of pore sizes in peat soils after the decomposition has started. The corresponding fractions of intra- and inner-plant pores, corresponding to the $|h|$ -ranges of 10–100 cm and above approximately 100 cm, respectively, can also be observed and distinguished in some of our WRF measurements. However, this bimodal structure is not reflected by the NMR RTDs. The resolution properties of the two NMR applications might not allow to distinguish between these pore clusters. This feature can be interpreted by a strong diffusive coupling of the two pore clusters, that is, the diffusing proton spins change their positions in the pore space during the NMR experiment so fast, that the corresponding relaxation regimes are averaged and appear as one broad mode in the RTDs.

On the other hand, the analogue of the WRF representing a

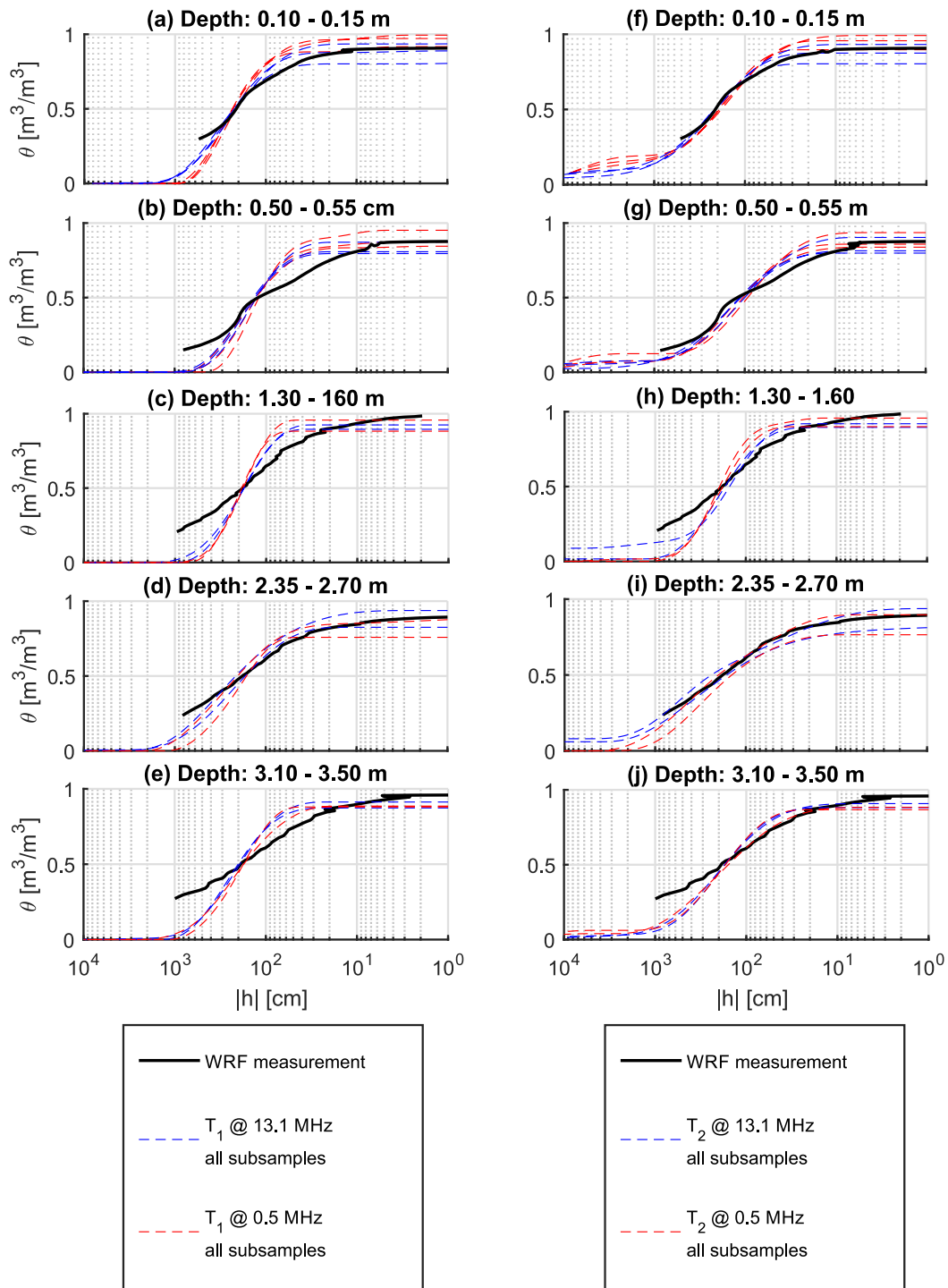


Fig. 7. Water retention functions (WRF) of five peat samples on sampling point P4 as measured using the evaporation method compared to NMR-based WRF representations of samples from the same depth interval as estimated from the cumulative T_1 (left hand side) and T_2 (right hand side) relaxation time distributions. For the undisturbed samples taken at depth < 60 cm, NMR data of three subsamples for each sampling depth has been acquired. For the repacked samples taken from regions > 60 cm, the NMR data of two subsamples for each depth interval has been acquired. The NMR measurements were conducted using the single-sided NMR (13.1 MHz) and the NMR core scanner (0.5 MHz).

cumulative pore size distribution can be questioned if anticipating the pore space of peat material being much more complex with angled polyhedrons, bottlenecks, dead end pores etc. The WRF is measured at different partial saturation states corresponding to various pore pressure heads and it is very likely that, instead of being simply stored in specific well-defined pore clusters, a significant part of the retained water is kept in menisci, capillary bridges, and even in large pores with openings far

too small for allowing air bubbles to enter. In contrast, the ^1H NMR signal is predominated by water-filled volumes and might actually represent the pore size distribution of peat material pretty well.

When comparing RTD and WRF, it must also be taken into account that the RTD measures the pore space in the saturated state, while the pores shrink during the WRF measurement. We observed a corresponding decrease of the peat sample volumes by up to 30 % during the

Table 1

Product of geometry factor and surface relaxivity $a \cdot \rho_1$ and $a \cdot \rho_2$ (see equation (9)) for the data depicted in Fig. 7.

		$a \cdot \rho_1$ [$\mu\text{m/s}$]	$a \cdot \rho_2$ [$\mu\text{m/s}$]	$a \cdot \rho_1$ [$\mu\text{m/s}$]	$a \cdot \rho_2$ [$\mu\text{m/s}$]
Sample depth [m]	Peat classification	(@13.1 MHz)	(@13.1 MHz)	(@0.5 MHz)	(@0.5 MHz)
0.10–0.15, A	amorphous peat	2.7	6.3	2.5	6.8
0.10–0.15, B	amorphous peat	2.3	4.6	2.3	6.3
0.10–0.15, C	amorphous peat	1.6	3.1	1.8	4.3
0.50–0.55, A	white	0.6	3.5	1.0	4.7
0.50–0.55, B	white	0.5	3.0	0.8	3.2
0.50–0.55, C	white	0.7	2.7	0.9	3.0
1.30–1.40	white	0.1	1.9	0.3	1.5
1.40–1.60	white	0.1	1.7	0.5	2.1
2.35–2.55	black	0.1	1.3	0.3	1.3
2.60–2.70	black	0.1	0.9	0.3	1.0
3.10–3.30	black	0.2	1.7	0.4	1.7
3.40–3.50	black	0.2	1.7	0.5	2.3

evaporation. Due to the forces that act during dewatering and cause the shrinkage of the pores, they dewater to a lesser extent than if they were rigid. In the rigid case, a correspondingly large pore would empty completely, whereas in the case of shrinkage, only the water volume corresponding to the shrinkage leaves the sample. The shrinking pores bind the water more strongly until it drains at higher suction stresses compared to an equivalent rigid pore with the same original size. According to Oleszczuk et al. (2003), it is even possible that the volume loss due to shrinkage can exceed that of the lost pore water, which is a strong indication for the reorganisation of the pore water during the desaturation. Already dewatered pores shrink further in the process and can be filled again at a certain size, if their increasing capillary force allows this. These effects may contribute to higher water contents at higher suction stresses that we observe in our WRF measurements, in contrast to the RTDs.

It is worth noting that the differences in the shapes between WRF and RTD could also be explained by a heterogeneous distribution of $\rho_{1/2}$ inside the pore space. The investigations of Jaeger et al. (2009) and Novotny et al. (2023) demonstrate that a homogeneous distribution of ρ_2 inside a soil sample cannot be considered to be the general case, especially in soils with high contents of fine material. A significant heterogeneity of the surface relaxivity, no matter if considering ρ_1 or ρ_2 , can corrupt the reconstruction of the WRF from NMR relaxation data unless calibration data is available that account for the different involved relaxation regimes. Both studies (Jaeger et al., 2009; Novotny et al., 2023) reported that the regimes with increased ρ_2 values are associated with decreased grain and correspondingly pore size.

4.5. Estimation and assessment of the NMR relaxation behaviour depending on the peat decomposition degree

Regardless of the open issues discussed in the previous subsection, we assume the existence of at least an effective pore size range represented by both WRF and RTD and, associated with the latter, effective ρ_1 and ρ_2 values for each sample. The fitting process described in Section 3.3, which transfers the RTD data into the h-domain, discovers estimates of $a \cdot \rho_1$ and $a \cdot \rho_2$. For the samples from the near surface, maximum values in the range of 1.6–2.7 $\mu\text{m/s}$ and 3.1–6.8 $\mu\text{m/s}$ for T_1 and T_2 , respectively, are observed. The values for $a \cdot \rho_{1/2}$ decrease with depth. The magnitude of this decrease depends on the kind of relaxation, i.e. T_1 or

T_2 , and on the frequency of the used device. At a depth of 2.35–2.65 m, the minimum values are reached with 0.1 $\mu\text{m/s}$ for the T_1 relaxation and about 1–2 $\mu\text{m/s}$ for T_2 . The difference between ρ_1 and ρ_2 is largest for the white peat samples and decreases with increasing decomposition. This effect represents the properties already discussed along with the T_1/T_2 ratios in Fig. 4.

The variation of $a \cdot \rho_1$ and $a \cdot \rho_2$ among the samples can be explained either by a depth-dependent change of $\rho_{1/2}$ and/or a change of the dominant pore geometry factor a . Considering the complete breakdown of the fibric material during the aerobic decomposition in the topsoil, one might expect a change of a from about 3 to 1, i.e. predominating circular pore geometry for the cellular structure of non-decomposed peat to predominating planar pore structure for the space remaining between the disintegrated organic fragments. However, because the difference of $a \cdot \rho_{1/2}$ among peat material with different decomposition degrees is larger than a factor of 3, we conclude that the alterations of the surface relaxivity generally outweigh the change of the pore geometry factor for both relaxation mechanisms.

It must be noted that the assumption of a zero contact angle ($\beta = 0$ in equation (9) for all of our samples is crucial as it directly affects the estimations discussed above. For the case that $\beta > 0$ we overestimate $a \cdot \rho_{1/2}$, which might be an issue for the decomposed peat from the topsoil. Gharedaghloo and Price (2019) measured β for decomposed peat up to 60° , a value that would scale our results in Table 1 by a factor of $\cos(60^\circ) = 0.5$. But even if we take this effect into account for the topsoil peat, we find that in particular the T_1 relaxation is still significantly increased compared to the poorly decomposed peat material from larger depths.

The study does not allow conclusions about any variations of $\rho_{1/2}$ inside the pore space of a single sample. Future NMR experiments to identify a heterogeneous $\rho_{1/2}$ distribution in a peat sample should be conducted during a stepwise dewatering of the peat samples at controlled pore pressure. An alternative measurement approach, referred to as Decay due to Diffusion in Internal Field (DDIF), might overcome at least the limitations given by a heterogeneous ρ_2 distribution inside the material under study (Novotny et al., 2023). However, this necessitates specific pulse sequences and measurement properties that are not available for the mobile NMR measurement equipment used in this study.

5. Conclusions and outlook

Our study demonstrates that NMR relaxometry is an effective tool to characterise the decomposition degree of peat material. The logarithmic mean relaxation time $T_{1,lgm}$ and $T_{2,lgm}$ are quantifiably correlated with common soil physical measures representing the decomposition state and might serve as corresponding proxies. The higher the degree of decomposition the faster is the NMR relaxation. This behaviour has been expected beforehand because the decomposition process leads to decreased pore dimensions (hypothesis 1).

$T_{1,lgm}$ and $T_{2,lgm}$ usually serve as robust and established proxies for the effective pore size in geophysical laboratory, borehole and field NMR applications and often allow the estimation of effective hydraulic parameters for mineral sediments. However, linking these two parameters to the hydraulic properties of peat soil remains an objective for future research. In this regard, it is important to concentrate future studies on the investigation of samples with largely unaltered hydraulic properties, i.e. samples as undisturbed as possible. This could be realised only in parts in this study and is difficult to achieve in general, particularly for peat samples.

We observe a significant dependence of the ratio between T_1 and T_2 from the peat decomposition degree, which indicates that the NMR relaxation behaviour does not only depend on pore space dimension. Highest values of more than ten are found for weakly decomposed peat, where the T_1/T_2 ratio increases with decreasing pore size. This is in

accordance to previous investigations and can be explained by the reduced mobility of the water molecules next to the pore surface (e.g. Brax et al., 2017), i.e. by their interactions with the polar molecular groups of the organic polymer network (hypothesis 2). With decreasing amount of polar groups due to decomposition, the T_1/T_2 ratios decrease remarkably. Our results suggest that the decomposition under aerobic conditions yields lower T_1/T_2 ratios than under anaerobic conditions. Future peat investigations using NMR relaxometry should thus involve T_1 and T_2 measurements to indicate both the changes of the pore sizes and the changes related to the physicochemical properties of the polymeric network. In doing so, attention must be paid to the fact that also the Larmor frequency of the used NMR relaxometry equipment affects the T_1/T_2 measurements. For mobile NMR equipment at low frequencies, the measurable contrasts in the T_1/T_2 ratios will be lower than those at higher frequencies. This is in particular true for borehole equipment, which works in a very similar frequency range as the mobile NMR scanner used in this study (Walsh et al., 2013; Mashhadi et al. 2024).

A detailed analysis of the influence of specific organic compounds in the peat material on the T_1/T_2 ratio must be the objective of future research. For this purpose, ^1H NMR relaxometry should ideally be combined with ^{13}C NMR spectroscopy, which is frequently used for analysing the components of organic soils (e.g. Serk et al., 2022).

Our study indicates that the pore-size related interpretation of the NMR RTDs is not straight forward. Therefore, our third hypothesis cannot be considered to be generally true. A certain variability of ρ_1 and ρ_2 must be expected in peat material depending on its individual characteristic decomposition state and correspondingly to its depth. Electron paramagnetic resonance data (EPR, e.g. Jezierski et al., 2008) for peat samples indicate that the electron spin density, the most likely source of paramagnetism in organic bog soils, can show strong variations along vertical soil profiles. NMR and EPR measurements should be combined in future peat studies to identify the mechanisms of the decomposition-based alterations of $\rho_{1/2}$.

CRedit authorship contribution statement

Stephan Costabel: Writing – review & editing, Writing – original draft, Visualization, Software, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Claus Florian Stange:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data has been made available for download from <https://doi.org/10.25928/80xb-g819>.

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